CS {4/6}290 & ECE {4/6}100 - Spring 2021

Homework 2: Caches

Dr. Thomas Conte

Due: February 12th, 2021 @ 11:59 PM

Rules

- **This is an individual assignment. ABSOLUTELY NO COLLABORATION IS PERMITTED**. All cases of honor code violations will be reported to the Dean of Students.
- The due date at the top of the assignment is final. Late assignments will not be accepted.
- Please use office hours for getting your questions answered. If you are unable to make it to office hours, please email the TAs.
- Make sure you **show your work and give explanations** for all questions unless otherwise stated.
- All answers must be **handwritten or typed**, and an electronic copy (PDF format) uploaded to Canvas before the due date.
- **Make sure your answers are written within the boxes we have provided. We will be using Gradescope for grading and answers outside the boxes will NOT be recognized by the system.**
- **There will be a point penalty for not following the above rule.**

Question 0 – Cache Warmup

(a) Assume a cache with (C = 10, B = 6, S = 0). Given that this cache implements a write-allocate, write-back policy, and system addresses are 32-bits wide:

(i) How many bits make up the *Tag*, *Index*, and *Block Offset* of an address accessing this Cache? (i.e. show the address split for accessing this cache)

(ii) How many bits are there in the *Data Store* of the Cache?

(iii) How many bits are there in the *Tag Store* of this Cache?

(b) Consider the C, B and S parameters of a Cache. What happens to compulsory, capacity and conflict misses if only the following parameter is changed (the other two are kept the same)?

(i) C is increased

(ii) S is increased

(iii) B is increased

Question 1 – Hand to Hand Caches

Consider the memory address trace below for a 32-bit address size system:

```
ST 0x0ECE4100
LD 0x0ECE6100
LD 0x00C1CD88
LD 0x0054B0FA
LD 0x00ABCD89
ST 0x00CDEF9A
LD 0x000CB017
LD 0x1002ED12
LD 0x1054B09B
LD 0x00ABCD88
LD 0x00EF7300
ST 0x000CB680
```

Hand simulate the above cache accesses for the *(C,B,S)* cache configurations (a)-(c). Assume that caches use the LRU replacement policy (where applicable) and implement the write-back, write-allocate scheme. **Answer questions (i)-(iii) for each of the configurations:**

(i) Indicate which accesses would cause a miss. If an access is a miss, identify the type of miss.
(ii) How many write-backs happen for this stream? **Explain why?**
(iii) If the cache policy was *write-through instead of write-back*, how many trips to main memory will there be for this stream? **Explain why?**

(a) (9, 4, 0)
(b) (9, 4, 4)
(c) (9, 4, 2)

Fill in this table to answer (i) {Hit : H, Miss : M} if (M) { Type}

Stream	(a)	(b)	(c)
ST 0x0ECE4100			
LD 0x0ECE6100			
LD 0x00C1CD88			
LD 0x0054B0FA			
LD 0x00ABCD89			
ST 0x00CDEF9A			
LD 0x000CB017			
LD 0x1002ED12			
LD 0x1054B09B			
LD 0x00ABCD88			
LD 0x00EF7300			
ST 0x000CB680			

Space for Answer to Question 1 (ii) and (iii)

Question 2 – Cache Design

Given the memory trace from problem 1 with parameter B = 4:

(a) What is the smallest direct-mapped cache for which no misses besides *compulsory* occur?
(b) What is the smallest Fully-Associative cache for which no misses besides *compulsory* occur?
(c) What is the smallest possible Set-Associative Cache (Not fully associative) that completely avoids *Capacity* and *Conflict* misses? (Find the minimum C and S values to satisfy the constraint)

Please show your work and explain

Space for Answer to Question 2

Question 3 – Cache Performance

A given memory hierarchy initially consists of an L1 cache and main memory. The L1 cache has a hit time of 2 cycles, and for a given workload a miss ratio (rate) of 15 percent. It takes 100 cycles to access main memory on average.

(a) What is the average access time for this system? **Show your work**

You are the chief memory architect for ACME computers and decide to improve your systems performance by considering the options listed in (b)-(d)

(b) Modification Option 1: Add a victim cache to the hierarchy. The victim cache has a miss ratio of 0.7. What is the average access time of the system now?

(c) Modification Option 2: Add a second level cache (L2) with the following characteristics: Hit Time = 8 Cycles, Miss Rate = 5 percent

(d) Modification Option 3: You give up improving your system cache entirely and instead want to tell your main memory supplier to make faster memory… Yikes! How fast must main memory be such that when used with the original system the overall AAT is less than 4 cycles? *(It is okay to have a fractional value for memory latency for this problem)*

(e) Which option do you choose after evaluating all of them or do you stick with your original memory system? **Please explain**